

YaP Link — native proxy (Velocity-class, phased)

Product decision: YaP Link is a **first-party Netty proxy** (`com.yapcore.link.*`), evolved from [yap-first-party/link/native/](#) . It is **not** a Velocity fork.

Status: Phases 0-6 in native Link (`0.6.0-phase6`).

See also: [yap-first-party/link/api/](#) · [yap-first-party/link/protocol/](#) · [yap-first-party/link/plugins/](#)

Velocity parity matrix

Feature	Velocity	YaP Link	Phase
Modern player-info forwarding	✓	✓	0 ✓
Multi-backend + try order	✓	✓ <code>link.properties</code> / <code>link.toml</code>	0 ✓
Online / offline login	✓	✓	0 ✓
Compression bridge	✓	✓	0 ✓
<code>/server</code> + Transfer reconnect	✓	✓	0 ✓
<code>link.properties</code> + <code>link.toml</code>	—	✓	0 ✓
Ping passthrough	✓	✓ cached backend probe	1 ✓
Forced hosts	✓	✓ <code>forced-host.<host>=server</code>	1 ✓
Backend health + try failover	✓	✓ <code>BackendMonitor</code>	1 ✓
Connect / login / read timeouts	✓	✓	1 ✓
Play-phase system chat	✓	✓ <code>PlayChat</code>	1 ✓
Aggregate player count in ping	✓	✓ <code>aggregate-player-count</code>	2 ✓
Cross-server chat relay	plugins	✓ <code>ChatRelay</code> + <code>say</code> console	2 ✓
Config reload	✓	✓ console <code>reload</code>	2 ✓
YaP Link plugin API	Velocity API	✓ <code>yap-link-api</code>	3 ✓
chat-bridge / mod-sync / server-selector	Velocity plugins	✓ <code>yap-first-party/link/plugins/</code>	3 ✓

Feature	Velocity	YaP Link	Phase
Bedrock UDP edge + per-backend routing	Geyser on proxy	✓ <code>BedrockUdpForwarder</code>	4 ✓
Floodgate key forwarding	Floodgate on proxy	✓ <code>floodgate-key.pem</code>	4 ✓
Metrics hooks	partial	✓ <code>LinkMetrics</code> + <code>/metrics</code>	5 ✓ / Edge ✓
Connect / handshake rate limit	plugins	✓ per-IP (default ON)	Edge ✓
<code>link-embed</code> in YaPcore	—	✓ <code>link-embed=true</code>	5 ✓
Release bundle	—	✓ <code>yap-link.jar</code> in <code>assembleRelease</code>	5 ✓
Velocity fork retired	—	✓ archived	5 ✓
Play plugin-message relay (<code>yap:chat</code>)	plugins	✓ <code>PluginMessagePackets</code>	6 ✓
Two-backend probe + chat relay	—	✓	6 ✓
Bedrock UDP forward to backend	—	✓	6 ✓

Phase 6 — Hardening

Task	Gate
Play-phase <code>PluginMessageEvent</code> for registered channels	Unit tests + chat-bridge relay
Multi-backend probe + <code>say</code> chat relay	
Bedrock UDP datagram forward to backend	

Config

Home: `link-data/` (or `--home`). Prefer `link.toml` or `link.properties` .

```
bind=0.0.0.0:25565
motd=YaP Link
servers.lobby=127.0.0.1:25566
servers.survival=127.0.0.1:25567
servers.lobby.bedrock=127.0.0.1:19132
try=lobby,survival
```

```
ping-passthrough=true
plugins-enabled=false
# First run: ./scripts/start-yap-link.sh seeds plugins-enabled=true + plugin jars
floodgate-key-file=floodgate-key.pem
bedrock-enabled=false
bedrock-bind=0.0.0.0:19132
bedrock-backend=127.0.0.1:19132
```

Plugins load from `link-data/plugins/*.jar` with `link-plugin.json` descriptors.

Build & run

```
gradle :yap-link-native:shadowJar
gradle :yap-link-plugin-chat-bridge:installIntoLinkPlugins # optional
./scripts/start-yap-link.sh
```

Embedded Link (dev / single-box): `config/server.properties`:

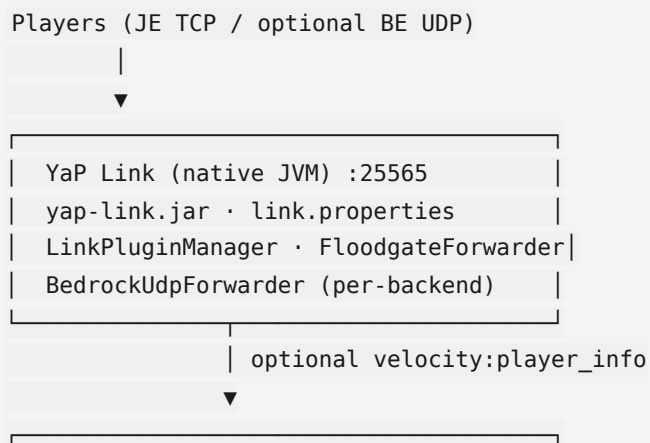
```
link-embed=true
link-embed-home=link-data
```

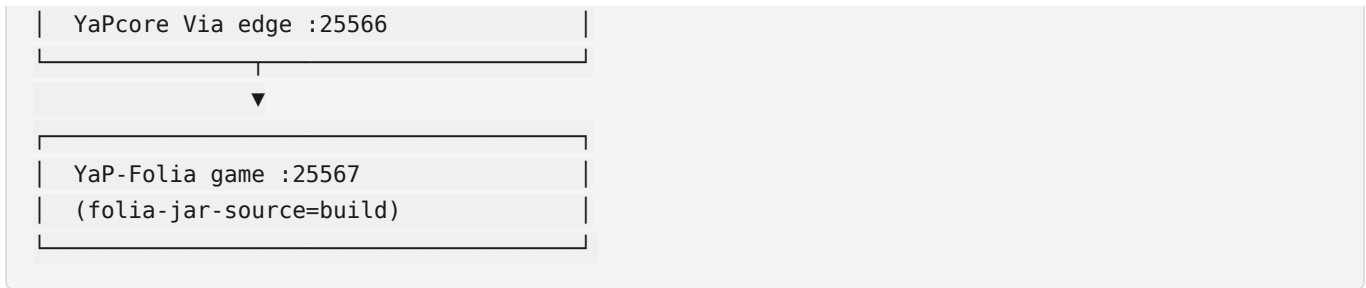
Requires `yap-link.jar` on the classpath alongside `yapcore.jar`.

Release: `gradle assembleRelease` ships `yapcore.jar`, `yap-link.jar`, and `link-data/plugins/`.

Console commands: `help` · `reload` · `list` · `servers` · `say <msg>` · `stop`

Architecture





Shared wire code: `yap-first-party/link/protocol/` (`McCodec` , `McFrameCodec` inbound, `McOutboundPacketEncoder` outbound `zlib+frame`, `McCompressionCodec` inbound `decompress + wrapOutbound` , modern forwarding, Floodgate).

Outbound rule: one handler for `zlib+length` (`McOutboundPacketEncoder`). Stacking `compress` `MessageToMessageEncoder` + frame encoder caused vanilla `CorruptedFrameException: Frame length cannot be zero after Set Compression`.

Login Success: if the backend never sends `velocity:player_info` (Folia velocity disabled), Link still bridges — forwarding is optional unless the backend requests it.

Jar the GUI starts: repo-root `yap-link.jar` (preferred) or `yap-first-party/link/native/build/libs/yap-link.jar` . Always republish/copy after protocol fixes so operators are not on a stale root jar.

Bedrock / Geyser

Bedrock UDP routing at Link is **shipped** (`BedrockUdpForwarder`) but **off by default** (`bedrock-enabled=false`). For single-box crossplay, use chassis dual-stack on the **YaP-Folia** backend. For BE-heavy multi-backend networks, enable Link Bedrock bind or use stock Velocity as a stand-in. See [YAP_LINK.md](#).

Agent handoff (Phases 0-2 ↔ 3-5)

For the 0-2 agent:

Rule	Detail
Plugins default OFF in code	<code>LinkConfig.applyDefaults()</code> → <code>plugins-enabled=false</code> . Unit tests and bare <code>LinkConfig.load()</code> stay plugin-free until config opts in.
First run / release turns plugins ON	<code>./scripts/start-yap-link.sh</code> seeds <code>link.properties</code> with <code>plugins-enabled=true</code> , builds <code>yap-link-plugin-*</code> jars into <code>link-data/plugins/</code> . <code>assembleRelease</code> should mirror that seed.

Rule	Detail
Bedrock ctor	<code>new BedrockUdpForwarder(LinkConfig)</code> only — not the old 4-arg host/port stub.
Plain chat	Built-in <code>ChatRelay</code> (Phase 2) — serverbound play chat + console <code>say</code> .
Backend <code>yap:chat</code>	<code>yap-link-plugin-chat-bridge</code> when backends send on channel <code>yap:chat</code> and <code>plugins-enabled=true</code> (<code>PluginMessageEvent</code>). Do not duplicate in <code>ClientSession</code> .

LinkServer owns: `LinkPluginManager`, `FloodgateForwarder`, `BedrockUdpForwarder(config)`, plus Phase 0-2 `BackendMonitor` / `ChatRelay` / `PlayerHub`.

Full play-phase plugin-message wire sniffing (all channels) is **optional future work** — not required for Phase 0-2 or for `yap:chat` when chat-bridge is loaded.

Boundary: Phase 0-2 stops at `ChatRelay`, `BackendMonitor`, `PlayChat`, console `reload`. Do not grow `ClientSession` into a second plugin platform.

Related

- [EDGE_RATE_LIMIT.md](#) · [EDGE_HARDEN.md](#) — rate limits, Prometheus, public edge harden
- [YAP_LINK.md](#) — operator entry
- [VELOCITY.md](#) — Folia backend forwarding
- [YAPCORE_WHITEPAPER.md](#) — chassis vs Link process boundaries